Practice workbook · Caesar cipher

Encrypting/decrypting with a Caesar shift, finding the shift, combining shifts, spotting wrap-around errors and explaining weakness.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-18

1. Question 4 · part a.i · 2 marks

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Worked-example retry · Attempt part a.i only.

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4 Data is encoded for encryption and for identifying records in databases.

(a) A Caesar cipher is one method of encryption.

(i) Complete the table to show the result of applying a Caesar cipher.

(2)

Plaintext	Shift	Ciphertext
PIXEL	−4	
CLOUD		FORXG

(ii) Plaintext is encrypted using a Caesar cipher.

Compare the ciphertexts produced for the plaintext **GOLD** by:

- a shift of −6 followed by a shift of +8
- a single shift of +2.

(2)

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(iii) Ann has written a Caesar cipher algorithm.

When the algorithm encrypts the plaintext **BYTE** with a shift of +5 the result is **GVYJ**.

This is incorrect.

Explain the error.

(2)

1. Question 4 · part a.i · 2 marks · Working and answer

Working and answer

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2. Question 3 · part b · 2 marks

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Further practice · Attempt part b only.

3 Several encryption algorithms have been developed.

(a) Ahmed has started to create a Caesar cipher encryption program.

Caesar cipher encryption works by giving a number value to a key.

Each letter in a plaintext message is replaced by a new ciphertext letter using the key.

For example, as shown in the table, a key value of +2 would change the plaintext message **encrypt** to the ciphertext message **gpetarv**.

Plaintext	e	n	c	r	y	p	t
Ciphertext	g	p	e	t	a	r	v

Ahmed's program already converts from plaintext to ciphertext.

He wants to extend the program to:

- accept the input of a plaintext message
- accept the input of the key, which must be a positive integer between 1 and 25
- write the generated ciphertext to a text file named **Cipher.txt**
- display the ciphertext.

Open **Q03a** in the code editor.

Complete the program code.

You must use the structure given in the file **Q03a** to complete the program.

Do not add any further functionality.

Save your code as **Q03aFINISHED** with the correct file extension for the programming language.

(8)

(b) Explain **one** reason why the Caesar cipher is a weak encryption algorithm.

(2)

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2. Question 3 · part b · 2 marks

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Further practice · Attempt part b only.

(c) The Vigenere cipher encrypts plaintext using a series of Caesar ciphers, based on the letters of a keyword and a Vigenere square.

Figure 1 shows a Vigenere square.

	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
b	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a
c	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b
d	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c
e	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d
f	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e
g	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f
h	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g
i	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h
j	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i
k	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j
l	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k
m	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l
n	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m
o	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n
p	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
q	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p
r	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q
s	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r
t	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s
u	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t
v	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u
w	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v
x	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w
y	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x
z	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y

Figure 1

Complete the table to perform a Vigenere decryption of the ciphertext message **flmktrwhee** using the keyword **thirty**.

(4)

Encrypted letter	f	l	m	k	t	r	w	h	e	e
Keyword letter										
Decrypted letter										

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P 6 6 4 9 0 A 0 9 1 6

9

Turn over ▶

2. Question 3 · part b · 2 marks · Working and answer

Working and answer

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3. Question 4 · part a.ii · 2 marks

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Further practice · Attempt part a.ii only.

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Working and answer

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4. Question 4 · part a.iii · 2 marks

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Further practice · Attempt part a.iii only.

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4. Question 4 · part a.iii · 2 marks

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Further practice · Attempt part a.iii only.

(b) A program is needed to create a key for a database.

A user enters a two-letter word, a whole number and a decimal number.

The program must ensure the word is only two characters long.

The program must display an error message when the word is not two characters long.

A key is generated from the whole number, the reversed word and the whole number part of the decimal number.

Figure 4 shows the input values and a valid key.

```
Enter a word: at
Enter a whole number: 12
Enter a decimal number: 7.89
12ta7
```

Figure 4

Open **Q04b** in the code editor.

Write the program.

Do not add any further functionality.

Save your code as **Q04bFINISHED** with the correct file extension for the programming language.

(8)

(Total for Question 4 = 14 marks)

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4. Question 4 · part a.iii · 2 marks · Working and answer

Working and answer

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